

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech.

Semester : VII

Branch: Electrical Engineering / Electrical Engineering (Electronics and Power)/ Electrical & Electronics Engg. / Electrical & Power Engineering.

Subject Code & Name: Power System Operation and Control (BTEEC701_Y20)

Max Marks: 60

Date:05/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	What is the main cause of voltage instability?	L1	1
	a. Generators b. Transformers c. Loads d. Line Losses		
	Reactive power is measured in ____	L1	1
	a. VA b. VAR c. W d. VAS		
3	Transmission losses in a line are ____	L1	1
	a. Directly b. Inversely c. Directly d. Inversely		
	Proportional to proportional to Proportional to proportional to		
	Voltage V Voltage V V^2 . V^2 .		
4	Which type of generator is used in Power System for absorbing Reactive Power?	L1	1
	a. Synchronous b. Diesel c. Static d. Brushless		
	Generator Generator Generator Generator		
	What is the primary objective of Load Frequency Control (LFC) in a power system?	L2	1
	a. To regulate b. To maintain c. To optimize d. To minimize		
	Voltage Levels a stable power flow transmission		
	frequency losses		

6	Which of the following is a characteristic of a brushless excitation system?				L2	1
	a. uses a separate DC generator	b. Uses a rotating rectifier	c. Uses a static power converter	d. Requires a separate exciter generator		
	Which of the following types of excitation systems is most commonly used in modern power plants?				L1	1
	a. Self excited	b. Separately excited	c. Brushless excitation	d. Static excitation		
8	Which of the following types of stability is concerned with the ability of the power system to return to its original state after a disturbance?				L2	1
	a. Steady state stability	b. Transient stability	c. Dynamic stability	d. Small signal stability		
9	What is the primary objective of economic operation of power system?				L1	1
	a. To minimize transmission losses	b. To maximize generation capacity	c. To minimize operating cost	d. To ensure reliable operation		
10	Which of the following methods is used to improve the voltage Stability of a power system?				L2	1
	a. Increasing the reactive power support	b. Reducing the transmission line impedance	c. Implementing automatic voltage regulators	d. Increasing the generation capacity		
11	What is the purpose of unit commitment in economic operation of a power system?				L2	1
	a. To determine the optimal generation schedule	b. To calculate the transmission losses	c. To evaluate the reliability of the system	d. To determine which generating units to commit to meet the load demand		

12	What is the main objective of optimal power flow (OPF) in economic operation of a power system?				L1	1
	a. To minimize transmission losses	b. To maximize generation capacity	c. To minimize operating cost	d. To optimize power flow in the transmission network		
Q.2	Solve the following.					12
A)	Explain the structure of electrical power system with neat diagram.				L2	6
B)	Explain the concepts of real power, reactive power and complex power.				L2	6
Q.3	Solve the following.					12
A)	Draw and explain schematic diagram of Synchronous machine with physical description.				L2	6
B)	Derive swing equation for a synchronous machine.				L3	6
Q.4	Solve Any Two of the following.					12
A)	Explain in detail elements of an excitation system.				L2	6
B)	List the various types of excitation system and Explain Dc Excitation system in detail.				L2	6
C)	Draw and explain functional block diagram of Power generation and control.				L2	6
Q.5	Solve Any Two of the following.					12
A)	Classify power system stability, Explain in detail.				L2	6
B)	Explain the concept of distribution of load between units within plants.				L3	6
C)	Write a short note on Unit Commitment.				L3	6

Q. 6	Solve Any Two of the following.		12
A)	Enlist the methods of Voltage Control and Explain any one of them.	L2	6
B)	Explain the Operation of a Synchronous Condenser with neat diagram.	L3	6
C)	Explain the construction and working of static VAR compensator.	L3	6
*** End ***			

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